Windows Defender Firewall

1. Create an Outbound Rule to Block Spotify

This rule prevents the Spotify desktop application from sending network traffic through Windows Firewall.

Steps

1. Press Win + R, type wf.msc, and press Enter.
2. Select Outbound Rules in the left panel.
3. Click New Rule... in the right panel.
4. Select Program, then click Next.
5. Select This program path and browse to the Spotify executable, commonly located in a path similar to:

C:\Users\YourName\AppData\Roaming\Spotify\Spotify.exe

The actual location may differ. You can find it through Spotify's shortcut properties or Task Manager.
6. Click Next and select Block the connection.
7. Select all profiles that apply: Domain, Private and Public.
8. Give the rule a name such as Block Spotify Outbound.
9. Click Finish.

Expected result: Spotify's network traffic should be blocked by this rule when Windows Firewall is active. Other applications should remain unaffected by this specific rule.

2. Create an Inbound Rule for Google Chrome on Port 443

Port 443 is the standard TCP port for HTTPS. Note that Chrome normally uses port 443 for outgoing connections to websites; it does not usually listen for incoming connections on local port 443. The following configures the requested inbound rule for a lab exercise.

Steps

1. Open Windows Defender Firewall with Advanced Security by running wf.msc.
2. Select Inbound Rules → New Rule....
3. Select Program and click Next.
4. Browse to the Google Chrome executable, usually:

C:\Program Files\Google\Chrome\Application\chrome.exe

Your installation path may differ.
5. Select Allow the connection.
6. Select the Domain, Private and Public profiles as appropriate.
7. Name the rule Allow Chrome HTTPS Inbound and finish creating it.
8. Open the rule's Properties → Protocols and Ports tab.

9. Set Protocol type to TCP and Local port to 443.
10. Click Apply and OK.

Important: This rule alone does not guarantee that only Chrome can receive connections on port 443. Other firewall rules and application behavior also matter. For normal Chrome web browsing, outbound HTTPS connections are the relevant traffic.

3. Block All Traffic to and From a Specific IP Address

Example IP address: 123.45.67.89

To block traffic in both directions, create an inbound rule and an outbound rule.

A. Block inbound traffic from the IP address

1. Open wf.msc.
2. Select Inbound Rules → New Rule....
3. Select Custom, then click Next.
4. Select All programs and click Next.
5. Leave the protocol setting as Any.
6. On the Scope page, under Remote IP address, select These IP addresses.
7. Click Add and enter 123.45.67.89.
8. Click Next, select Block the connection, and continue.
9. Select all applicable profiles.
10. Name the rule Block Inbound 123.45.67.89 and click Finish.

B. Block outbound traffic to the IP address

Repeat the same process under Outbound Rules, using the same remote IP address and selecting Block the connection.

Name the rule Block Outbound 123.45.67.89.

Result: Windows Firewall will block matching inbound and outbound traffic involving that remote IP address, subject to the applicable firewall profiles and rule configuration.

4. Hardware Firewall vs. Software Firewall

Feature	**Hardware Firewall**	**Software Firewall**
Location	A dedicated physical device or network appliance.	Runs on a computer or server.
Protection	Can protect an entire network at its boundary.	Can filter traffic for an individual device.

Management	Often centrally managed by network administrators.	Managed on each computer or through centralized policies.
Example use case	A large office protecting its LAN from external threats.	Windows Defender Firewall protecting a laptop on home Wi-Fi.

Three key differences

1. Deployment: Hardware firewalls sit between networks; software firewalls run on individual devices.
2. Coverage: Hardware firewalls can protect many devices at once, while a software firewall generally protects the host where it runs.
3. Resources and cost: Hardware firewalls require appliance equipment and maintenance; software firewalls use the computer's resources and are often built into the operating system.

Conclusion: Hardware firewalls are useful for protecting an organization's network, while software firewalls provide device-level protection. Many organizations use both for layered security.